

SLIDRONIX

Phase C → Phase E Complete Technical Documentation

AI-Based Landslide Early Warning System

Western Ghats Study Region • Environmental Dataset Construction

Phase	Status	Main output
Phase C	Completed	NDVI + Land Cover + Soil for 14,270 samples
Phase D	Completed	CHIRPS reference-period rainfall features
Phase E	Completed	Integrated landslide/background dataset

1. Purpose and roadmap position

This document records the actual implementation completed in Phases C, D and E. It is intended for project documentation and for explaining the work to team members.

Roadmap: A → B → C → D → E → F → G → H

This document deliberately stops at Phase E. CNN, Transformer, fusion modelling, model comparison and risk mapping belong to later phases.

2. Phase C — Satellite & Soil Features

Phase C added remotely sensed vegetation, land-cover and soil information to the terrain representation. All satellite and soil processing was performed for the 14,270-sample clean modelling set.

2.1 NDVI — MODIS MOD13Q1 Version 6.1

NDVI was obtained through NASA AppEEARS using MOD13Q1.061 at 250 m with 16-day composites. The 14,270 points were submitted in 15 batches because the point workflow was split into batches of up to 1,000 coordinates.

Reference year: 2022. The 2021-12-19 observation was excluded so the annual features represent 2022 observations.

Pixel reliability was used as QA. Reliability 0 (good) and 1 (marginal) were retained; other reliability classes were excluded from the annual NDVI aggregation.

Processing correction: the downloaded AppEEARS values were already in the normal NDVI range. An initial script multiplied them by 0.0001 a second time, producing values around 0.00007. The mistake was detected from the summary statistics and corrected. The final pipeline retains the AppEEARS NDVI values directly.

Feature	Meaning
NDVI_mean	Mean valid 2022 NDVI
NDVI_std	Temporal variability
NDVI_min	Minimum valid NDVI
NDVI_max	Maximum valid NDVI
NDVI_range	Maximum minus minimum NDVI
NDVI_valid_observations	Count of valid QA-filtered observations

Final NDVI processing covered 14,270 unique IDs. Seventeen samples had no valid NDVI after QA filtering; these remain missing rather than being fabricated.

2.2 Land Cover — MODIS MCD12Q1 Version 6.1

Annual land cover was extracted using MCD12Q1.061 LC_Type1 at 500 m. All 14,270 samples received a land-cover class and QC value.

LC_Type1	Count
1	2
2	2,837
4	277
5	142
6	2
8	4,332
9	1,864
10	469
11	62
12	3,252
13	109
14	874
16	2
17	46

2.3 Soil — SoilGrids

Soil properties were extracted from SoilGrids at 250 m for the 0–5 cm depth interval using the median estimate (Q0.5). Every one of the 14,270 samples had valid values for all six properties.

Feature	Property / unit
Soil_clay_0_5cm	Clay content (%)
Soil_sand_0_5cm	Sand content (%)
Soil_silt_0_5cm	Silt content (%)
Soil_soc_0_5cm	Soil organic carbon (g/kg)
Soil_bdod_0_5cm	Bulk density (g/cm ³)
Soil_phh2o_0_5cm	Soil pH

3. Phase D — Rainfall Features

Rainfall was implemented using CHIRPS v3 daily final satellite rainfall. To keep the prototype computationally practical, the frozen reference period is 1–30 January 2022.

Interpretation: these are reference-period rainfall features, not event-aligned pre-landslide rainfall triggers and not a real-time rainfall feed.

Feature	Description
rainfall_total_30d_mm	30-day accumulated rainfall
rainfall_mean_daily_mm	Mean daily rainfall
rainfall_max_daily_mm	Maximum daily rainfall
rainfall_std_daily_mm	Daily rainfall standard deviation
rainfall_7d_mm	7-day accumulation
rainfall_15d_mm	15-day accumulation
rainfall_30d_mm	30-day accumulation
rainfall_max_7d_mm	Maximum 7-day accumulation
rainfall_max_15d_mm	Maximum 15-day accumulation
rainfall_max_30d_mm	Maximum 30-day accumulation

The rainfall feature dataset contains 14,270 samples with no missing rainfall feature values.

4. Phase E — Dataset Expansion

Phase E converts the historical inventory into a binary landslide/background modelling dataset. Positive samples are historical landslide locations; background samples are locations sampled within the project study region.

4.1 Study region and background sampling

The study boundary was derived from administrative boundaries and a spatial envelope around the historical landslide inventory. It should be described as a **Western Ghats study boundary derived from administrative boundaries and the historical landslide inventory**, not as an official authoritative Western Ghats boundary.

The original binary dataset contained 7,885 positive landslide records and 7,885 background records. Background sampling was constrained to the study polygon and used a minimum geodesic separation of approximately 1 km from positive locations.

4.2 DEM filtering and final sample set

DEM extraction produced elevation and slope for 14,270 of the 15,770 initial binary samples. 1,500 samples without valid DEM coverage after the defined repair/diagnostic process were removed rather than assigning invented terrain values.

The final clean integrated dataset contains **14,270 samples: 6,991 landslide and 7,279 background**. It is therefore close to balanced but not exactly 50:50 after DEM-validity filtering.

4.3 Integrated feature dataset

Component	Samples	Features
Terrain	14,270	Elevation, slope
Satellite	14,270	NDVI statistics, land cover
Soil	14,270	Clay, sand, silt, SOC, bulk density, pH
Rainfall	14,270	10 reference-period rainfall features

Component	Samples	Features
Target	14,270	Landslide = 1; Background = 0

The final integrated file contains 40 columns in total and 24 explicitly identified modelling features: 2 terrain + 6 satellite + 6 soil + 10 rainfall. The target is the binary Landslide field.

5. Data-quality checks

- 14,270 environmental rows and 14,270 AppEEARS point records were verified.
- All 14,270 environmental/AppEEARS coordinates matched after rounding to six decimal places.
- 14,270 unique sample IDs were verified.
- All six soil variables had 14,270 valid values.
- All rainfall features had 14,270 valid values.
- All land-cover records were present.
- NDVI had only 17 samples without any valid observation after QA.
- Repeated coordinates were preserved rather than automatically deleting historical records.
- Missing DEM values were not fabricated; affected samples were removed from the clean modelling set.

6. Important limitations

1. Rainfall currently uses a fixed January 2022 reference period, not event-aligned rainfall sequences.
2. Background points are modelling samples and do not prove that a location has never experienced a landslide.
3. The study boundary is project-derived and should not be presented as an official Western Ghats boundary.
4. Seventeen samples have no valid NDVI after QA filtering and will require appropriate model-side missing-value handling.
5. Phase F will determine how static environmental features and rainfall information are represented for the deep-learning architecture.

7. Files produced

File	Purpose
data/processed/ndvi_2022_features_14270.csv	Annual NDVI features
data/processed/land_cover_2022_features_14270.csv	Annual land-cover features
data/processed/satellite_features_14270.csv	NDVI + land-cover features
data/processed/soil_features_14270.csv	SoilGrids features
data/processed/rainfall/chirps_2022_30day_features_14270.csv	CHIRPS rainfall features
data/processed/slidronix_integrated_dataset_14270.csv	Integrated Phase C–E dataset

8. Phase C–E completion checkpoint

Phase C: completed. **Phase D:** completed. **Phase E:** integrated dataset completed with 14,270 samples and 24 modelling features. The next roadmap stage is Phase F.